

Linear Structure

1. Introduction
2. Stacks
3. Queues
4. Deques

3.1~3.2 What Are Linear ADT?

They are data collections whose items are **ordered** depending on how they are added or removed. Once an item is added, it stays in that position relative to the other elements that came before and came after it. Collections such as these are often referred to as linear ADT.

Linear ADT can be thought of as having two ends. Sometimes these ends are referred to as the left and the right, or in some cases the front and the rear.

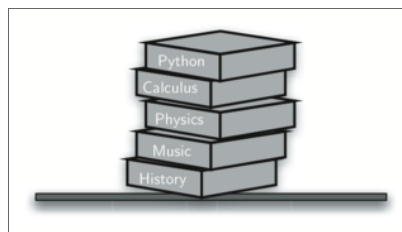
What distinguishes one linear structure from another is the way in which items are added and removed, in particular the **location where these additions and removals occur**.

3.3 Stacks

A stack (sometimes called a push-down stack) is an ordered collection of items where the addition of new items and the removal of existing items always takes place **at the same end**. This end is commonly referred to as the top. The end opposite the top is known as the base.

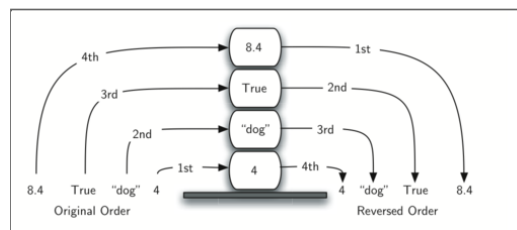
The base of the stack is significant since items stored in the stack that are closer to the base represent those that have been in the stack the longest. The most recently added item is the one that is in position to be removed first. This ordering principle is sometimes called LIFO, or last in, first out.

Almost any cafeteria has a stack of trays or plates where you take the one at the top, uncovering a new tray or plate for the next customer in line. Imagine a stack of books on a desk. The only book whose cover is visible is the one on top. To access others in the stack, we need to remove the ones that are sitting on top of them.



Assume you start out with a clean desktop. Now place books one at a time on top of each other. You are constructing a stack. Consider what happens when you begin removing books. The order that they are removed is exactly the reverse of the order that they were placed.

Stacks are fundamentally important, as they can be used **to reverse the order of items**. The order of insertion is the reverse of the order of removal:



For example, every web browser has a Back button. As you navigate from web page to web page, those pages are placed on a stack (actually it is the URLs that are going on the stack). The current page that you are viewing is on the top and the first page you looked at is at the base. If you click on the Back button, you begin to move in reverse order through the pages!

3.4 The Stack Abstract Data Type

The stack operations are given below:

- `Stack()` creates a new stack that is empty. It needs no parameters and returns an empty stack.
- `push(item)` adds at the top and returns `void`.
- `top()` reads the top without removing it.
- `pop()` removes the top and returns `void`; call `top()` first when the value is needed.
- `empty()` tests whether the stack is empty.
- `size()` returns the number of elements.

Calling `top()` or `pop()` on an empty `std::stack` violates its precondition.

For `std::stack`, only the top is observable; the conceptual table lists the top item at the far right.

<code>std::stack<int></code> operation	Conceptual contents (top at right)	Return value
<code>s.empty()</code>	<code>[]</code>	<code>true</code>
<code>s.push(4)</code>	<code>[4]</code>	<code>void</code>
<code>s.push(7)</code>	<code>[4, 7]</code>	<code>void</code>
<code>s.top()</code>	<code>[4, 7]</code>	<code>7</code>
<code>s.pop()</code>	<code>[4]</code>	<code>void</code>
<code>s.size()</code>	<code>[4]</code>	<code>1</code>

3.5 Using `std::stack` in C++

The cppds C++ text uses the Standard Library container adapter `std::stack<T>`. Its public operations are `push`, `pop`, `top`, `empty`, and `size`. Unlike the course's teaching class, STL `pop()` removes an item but returns `void`; read `top()` before popping when the value is needed.

`std::stack` is a **container adapter**: by default it stores elements in a `std::deque`, while exposing only LIFO operations. Client code depends on the stack contract rather than on an arbitrary position in the underlying container.

STL mapping: `push(item)` adds at the top, `top()` observes the top, `pop()` removes it, and `empty()` / `size()` inspect the state. Calling `top()` or `pop()` on an empty stack violates the precondition.

The following is the official C++ interface. A stack object has one element type, such as `stack<string>`:

```
#include <stack>

std::stack<std::string> s;
s.push("first");
s.push("second");
std::cout << s.top();    // second
s.pop();                 // removes second; returns void
```

Three distinct APIs are used in this course:

API	Construction	Inspect top	Remove top
STL <code>std::stack<T></code>	<code>std::stack<T> s;</code>	<code>top()</code>	<code>pop()</code> returns void
python3 <code>Stack<T></code>	<code>Stack<T> s;</code>	<code>peek()</code>	<code>pop()</code> returns T
HW3 capacity stack	<code>Stack<T> s(capacity);</code>	<code>stackTop()</code> or <code>peek(index)</code>	<code>pop()</code> returns T

The HW3 class additionally provides `isFull()`. Do not mix method names between these interfaces.

First use the canonical STL adapter. The following cell is self-contained and shows its value-before-pop pattern.

In [1]:

```
#include <iostream>
#include <stack>
#include <string>
using namespace std;

int main() {
    stack<string> s;
    s.push("4"); s.push("dog"); s.push("true");
    cout << s.size() << endl;
    cout << boolalpha << s.empty() << endl;
    cout << s.top() << endl;
    s.pop();
    cout << s.top() << endl;
}
```

3

false

true

dog

We could instead keep the **top** at the beginning of the **vector**:

```
template <typename T>
class Stack2 {
    private:
        vector<T> items;

    public:
        bool isEmpty() {
            return items.empty();
        }
        void push(T item) {
            items.insert(items.begin(), item);    // O(n)!
        }
        T pop() {
            T top = items.front();
            items.erase(items.begin());    // O(n)!
            return top;
        }
        T peek() {
            return items.front();
        }
        int size() {
            return items.size();
        }
};
```

In this case, `push_back()` and `pop_back()` no longer touch the top, so we use `insert(begin())` and `erase(begin())` instead — every push and pop now shifts all elements, turning $O(1)$ operations into $O(n)$.

In [2]:

```
#include <iostream>
#include "pythonds3/cppds/stack.hpp"    // Stack2: top at the front,  $O(n)$  ops
using namespace std;

int main() {
    Stack2<double> s;
    cout << boolalpha << s.isEmpty() << endl;
    s.push(4);
    s.push(7.5);
    cout << s.peek() << endl;
    s.push(2);
    cout << s.size() << endl;
    cout << s.pop() << endl;
    cout << s.pop() << endl;
    cout << s.size() << endl;
    return 0;
}
```

true

7.5

3

2

7.5

1

This ability to change the physical implementation of an ADT while maintaining the logical characteristics is an example of abstraction at work. However, even though the stack will work either way, if we consider the performance of the two implementations, there is definitely a difference!

For `std::stack`, `push`, `pop`, and `top` are constant time with the default `deque` backend. For the supplementary vector-backed class, tail `push_back` is amortized $O(1)$ and tail `pop_back` is $O(1)$.

The supplementary `Stack2` deliberately puts the top at `vector::begin()`; both insertion and erasure then shift elements and cost $O(n)$. It illustrates that identical ADT behavior does not imply identical implementation cost.

In []:

```
display_quiz(path+"stack1.json", max_width=800)
```

Given the following sequence of stack operations, what is the top item on the stack when the sequence is complete?

```
std::stack<std::string> m;  
m.push("x");  
m.push("y");  
m.pop();  
m.push("z");  
m.top();
```

☐ "z"

☐ "y"

☐ "x"

☐ The stack is empty

In []:

```
display_quiz(path+"stack2.json", max_width=800)
```

What happens when this std::stack code executes?

```
std::stack<std::string> m;  
m.push("x");  
m.push("y");  
m.push("z");  
while (!m.empty()) {  
    m.pop();  
    m.pop();  
}
```

☐ a pop on an empty stack occurs (undefined behavior / likely crash)

☐ "z"

☐ the stack is empty

☐ "x"

EXERCISE: WRITE `revString(const string&)` **USING**
`std::stack<char>`.

In [3]:

```
#include <iostream>
#include <stack>
using namespace std;

string revString(string myStr) {
    stack<char> s;
    // Step 1: push every char onto s; Step 2: pop to build rStr
    string rStr = "";
    while (!s.empty()) {
        rStr =
    }
    return rStr;
}

int main() {
    cout << revString("NSYSU") << endl;
    return 0;
}
```

```
/tmp/tmpo1rvkctj.cpp: In function 'std::string revString(st
d::string)':
/tmp/tmpo1rvkctj.cpp:13:5: error: expected primary-expressi
on before '}' token
```

```
13 |     }
    |     ^
```

```
[C++ kernel] Error: Unable to compile the source code. Retu
rn error: 0x1.
```

Expected output: `USYSN`

3.8. Converting Decimal Numbers to Binary Numbers

Binary representation is important in computer science since all values stored within a computer exist as a string of binary digits, a string of 0s and 1s.

Integer values are common data items. They are used in computer programs and computation all the time. We learn about them in math class and of course represent them using the decimal number system, or base 10. The decimal number 233_{10} and its corresponding binary equivalent 11101001_2 are interpreted respectively as:

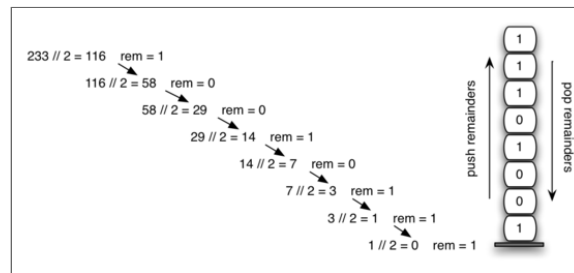
$$2 \times 10^2 + 3 \times 10^1 + 3 \times 10^0$$

$$1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

But how can we easily convert integer values into binary numbers? The answer is an algorithm called Divide by 2 that uses a stack to keep track of the digits for the binary result!

The Divide by 2 algorithm assumes that we start with an integer greater than 0. A simple iteration then continually divides the decimal number by 2 and keeps track of the remainder.

The first division by 2 gives information as to whether the value is even or odd. An even value will have the digit 0 in the ones place. An odd value will have the digit 1 in the ones place. We think about building our binary number as a sequence of digits; **the first remainder we compute will actually be the last digit in the sequence**



We again see the reversal property that signals that a stack is likely to be the appropriate data structure for solving the problem!

The C++ code below implements the Divide by 2 algorithm. The function `divide_by_2()` takes an argument that is a decimal number and repeatedly divides it by 2:

In [6]:

```
#include <iostream>
#include <stack>
#include <string>
using namespace std;

string divideBy2(int decimalNum) {
    stack<int> remStack;
    while (decimalNum > 0) {
        remStack.push(decimalNum % 2);
        decimalNum = decimalNum / 2;
    }
    string binString = "";
    while (!remStack.empty()) {
        binString += to_string(remStack.top());
        remStack.pop();
    }
    return binString;
}

int main() {
    cout << divideBy2(42) << " " << divideBy2(31) << endl;
    return 0;
}
```

101010 11111

Expected: 101010 11111.

It can easily be extended to perform the conversion for any base. In computer science it is common to use a number of different encodings. The most common of these are binary, octal (base 8), and hexadecimal (base 16).

The decimal number 233_{10} and its corresponding octal and hexadecimal equivalents 351_8 and $E9_{16}$ are interpreted as

$$3 \times 8^2 + 5 \times 8^1 + 1 \times 8^0$$

$$14 \times 16^1 + 9 \times 16^0$$

The function `divide_by_2()` can be modified to accept not only a decimal value but also a base for the intended conversion. The "Divide by 2" idea is simply replaced with a more general "Divide by base."

Base 2 through base 10 numbers need a maximum of 10 digits, so the typical digit characters 0 through 9 work fine.

The problem comes when we go beyond base 10. We can no longer simply use the remainders, as they are themselves represented as two-digit decimal numbers. Instead we need to create a set of digits that can be used to represent those remainders beyond 9

In [7]:

```
#include <iostream>
#include <stack>
using namespace std;

string baseConverter(int decimalNum, int base) {
    string digits = "0123456789ABCDEF";
    stack<int> remStack;
    while (decimalNum > 0) {
        remStack.push(decimalNum % base);    decimalNum /= base;
    }
    string newString = "";
    while (!remStack.empty()) {
        newString += digits[remStack.top()];    remStack.pop();
    }
    return newString;
}

int main() {
    cout << baseConverter(25, 2) << " " << baseConverter(25, 16) << endl;
    return 0;
}
```

11001 19

Expected: 11001 19 — the same value in binary and hexadecimal.

In []:

```
display_quiz(path+"convert.json", max_width=800)
```

What is the value of 25 expressed as an octal number?

Type your answer here:

3.9. Infix, Prefix, and Postfix Expressions

When you write an arithmetic expression such as $B \cdot C$, the form of the expression provides you with information so that you can interpret it correctly.

In this case we know that the variable B is being multiplied by the variable C since the multiplication operator $\cdot$ appears between them in the expression. This type of notation is referred to as infix since the operator is **in between** the two operands that it is working on.

Consider another infix example, $A + B \cdot C$. The operators $+$ and $\cdot$ still appear between the operands, but there is a problem. Which operands do they work on?

Does the $+$ work on A and B , or does the $\cdot$ take B and C ? The expression seems ambiguous. In fact, we know something about the operators $+$ and $\cdot$. Each operator has a precedence level. Operators of higher precedence are used before operators of lower precedence.

The only thing that can change that order is the presence of parentheses. The precedence order for arithmetic operators places multiplication and division above addition and subtraction. If two operators of equal precedence appear, then a left-to-right ordering or associativity is used.

Let's interpret the troublesome expression $A + B \cdot C$ using operator precedence. B and C are multiplied first, and A is then added to that result. $(A + B) \cdot C$ would force the addition of A and B to be done first before the multiplication. In the expression $A + B + C$, by precedence (via associativity), the leftmost $+$ would be done first.

Computers need to know exactly what operations to perform and in what order.

One way to write an expression that guarantees there will be no confusion with respect to the order of operations is to create what is called a fully parenthesized expression. This type of expression uses one pair of parentheses for each operator. The parentheses dictate the order of operations; there is no ambiguity!

The expression $A + B \cdot C + D$ can be rewritten as $((A + (B \cdot C)) + D)$ to show that the multiplication happens first, followed by the leftmost addition. $A + B + C + D$ can be written as $((A + B) + C) + D$ since the addition operations associate from left to right.

There are two other very important expression formats that may not seem obvious to you at first. Consider the infix expression $A + B$. What would happen if we moved the operator before the two operands? The resulting expression would be $+ A B$. Likewise, we could move the operator to the end, resulting in $A B +$. These look a bit strange.

These changes to the position of the operator with respect to the operands create two new expression formats, prefix and postfix. Prefix expression notation requires that all operators precede the two operands that they work on. Postfix, on the other hand, requires that its operators come after the corresponding operands.

$A + B \cdot C$ would be written as $+ A \cdot B C$ in prefix. The multiplication operator comes immediately before the operands B and C , denoting that $\cdot$ has precedence over $+$. The addition operator then appears before the A and the result of the multiplication.

In postfix, the expression would be $A\ B\ C\ \cdot\ +$. Again, the order of operations is preserved since the $\cdot$ appears immediately after the B and the C , denoting that $\cdot$ has precedence, with $+$ coming after.

Now consider the infix expression $(A + B) \cdot C$. Recall that in this case, infix requires the parentheses to force the performance of the addition before the multiplication.

However, when $A + B$ was written in prefix, the addition operator was simply moved before the operands, $+ A B$. The result of this operation becomes the first operand for the multiplication. The multiplication operator is moved in front of the entire expression, giving us $\cdot + A B C$. Likewise, in postfix $A B +$ forces the addition to happen first:

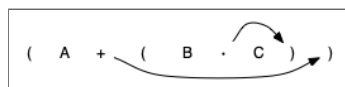
Infix Expression	Prefix Expression	Postfix Expression
$A + B$	$+ A B$	$A B +$
$A + B \cdot C$	$+ A \cdot B C$	$A B C \cdot +$
$(A + B) \cdot C$	$\cdot + A B C$	$A B + C \cdot$
$A + B \cdot C + D$	$+ + A \cdot B C$ D	$A B C \cdot + D$ +
$(A + B) \cdot (C + D)$	$\cdot + A B + C$ D	$A B + C D +$ $\cdot$
$A \cdot B + C \cdot D$	$+ \cdot A B \cdot C$ D	$A B \cdot C D \cdot$ +
$A + B + C + D$	$+ + + A B C$ D	$A B + C + D$ +

3.9.1 Conversion of Infix Expressions to Prefix and Postfix

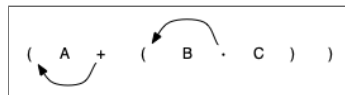
The first technique that we will consider uses the notion of a fully parenthesized expression that was discussed earlier.

Recall that $A + B \cdot C$ can be written as $(A + (B \cdot C))$ to show explicitly that the multiplication has precedence over the addition. Look at the **right parenthesis** in the subexpression $(B \cdot C)$ above. If we were to move the multiplication symbol to that position and remove the matching left parenthesis, giving us $B C \cdot$, we would in effect have converted the subexpression to postfix notation.

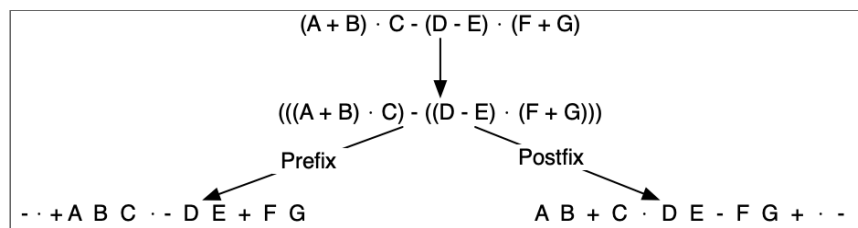
If the addition operator were also moved to its corresponding right parenthesis position and the matching left parenthesis were removed, the complete postfix expression would result:



If we do the same thing but instead of moving the symbol to the position of the right, we move it to the **left parenthesis**, we get prefix notation. The position of the parenthesis pair is actually a clue to the final position of the enclosed operator.



So in order to convert an expression, no matter how complex, to either prefix or postfix notation, fully parenthesize the expression using the order of operations. Then move the enclosed operator to the position of either the left or the right parenthesis depending on whether you want prefix or postfix notation:



3.9.2. General Infix-to-Postfix Conversion

Consider once again the expression $A + B \cdot C$. As shown above, $A B C \cdot +$ is the postfix equivalent. We have already noted that the operands A , B , and C stay in their relative positions. It is only the operators that change position.

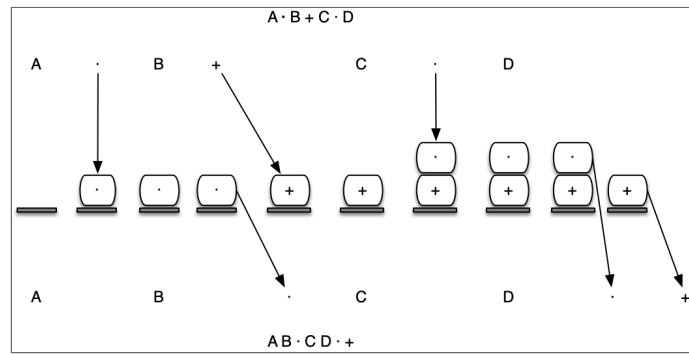
Let's look again at the operators in the infix expression. The first operator that appears from left to right is $+$. However, in the postfix expression, $+$ is at the end since the next operator, $\cdot$, has precedence over addition. **The order of the operators in the original expression is reversed** in the resulting postfix expression.

As we process the expression, the operators have to be saved somewhere since their corresponding right operands are not seen yet. Also, the order of these saved operators may need to be reversed due to their precedence.

Because of this reversal of order, it makes sense to consider using a stack to keep the operators until they are needed! What about $(A + B) \cdot C$? Recall that $A B + C \cdot$ is the postfix equivalent. Again, processing this infix expression from left to right, we see $+$ first. In this case, when we see $\cdot$, $+$ has already been placed in the result expression because it has precedence over $\cdot$ by virtue of the parentheses.

The conversion algorithm works by saving a left parenthesis to indicate the upcoming arrival of a high-precedence operator, which waits for the corresponding right parenthesis to determine its position. When the right parenthesis appears, the operator is popped from the stack.

1. Create `std::stack<string> operators` and `vector<string> output`.
2. Tokenize the input with `std::istringstream`.
3. Scan the token list from left to right:
 - Append operands to the output list.
 - Push left parentheses onto `op_stack`.
 - For right parentheses, pop `op_stack` until the left parenthesis is removed, appending each operator to the output list.
 - Push operators (`.`, `/`, `+`, `-`) onto `op_stack` after removing any higher or equal precedence operators from the stack and appending them to the output list.
4. After processing the input expression, remove any remaining operators from `op_stack` and append them to the output list.



Tokens are extracted with `std::istringstream`; the complete C++ program is below.

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```
In [8]: #include <iostream>
#include "pythonds3/cppds/expression.hpp" // the complete converter (md list
using namespace std;

int main() {
    cout << infixToPostfix("A * B + C * D") << endl;
    cout << infixToPostfix("( A + B ) * C - ( D - E ) * ( F + G )") << endl;
    return 0;
}
```

AB*CD*+

AB + C * DE - FG + * -

Expected:

```
A B * C D * +  
A B + C * D E - F G + * -
```

3.9.3. Postfix Evaluation

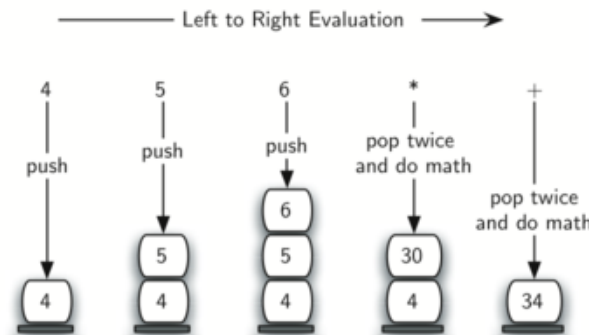
We will now consider the evaluation of an expression that is already in postfix notation. In this case, a stack is again the data structure of choice. However, as you scan the postfix expression, **it is the operands that must wait, not the operators as in the conversion algorithm above.**

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Another way to think about the solution is that whenever an operator is seen on the input, the two most recent operands will be used in the evaluation.

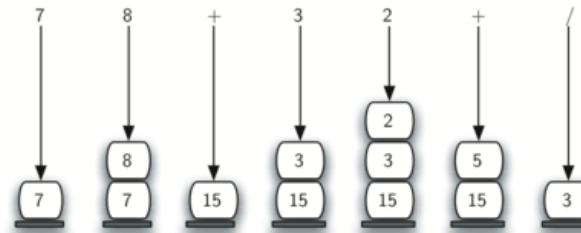
To see this in more detail, consider the postfix expression `4 5 6 · +`. As you scan the expression from left to right, you first encounter the operands `4` and `5`. At this point, you are still unsure what to do with them until you see the next symbol. Placing each on the stack ensures that they are available if an operator comes next.

To see this in more detail, consider the postfix expression `4 5 6 · +`. As you scan the expression from left to right, you first encounter the operands `4` and `5`. At this point, you are still unsure what to do with them until you see the next symbol. Placing each on the stack ensures that they are available if an operator comes next.



A slightly more complex example, `7 8 + 3 2 + /`. First, the stack size grows, shrinks, and then grows again as the subexpressions are evaluated. Second, the division operation needs to be handled carefully. Recall that the operands in the postfix expression are in their original order since postfix changes only the placement of operators. When the operands for the division are popped from the stack, they are reversed. Since division is not a commutative operator, we must be sure that the order of the operands is not switched!

A slightly more complex example, `7 8 + 3 2 + /`. First, the stack size grows, shrinks, and then grows again as the subexpressions are evaluated. Second, the division operation needs to be handled carefully. Recall that the operands in the postfix expression are in their original order since postfix changes only the placement of operators. When the operands for the division are popped from the stack, they are reversed. Since division is not a commutative operator, we must be sure that the order of the operands is not switched!



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2. Tokenize with `std::istringstream` and scan from left to right.
3. Scan the token list from left to right.
 - If the token is an operand, convert it from a string to an integer and push the value onto the `operand_stack`.
 - If the token is an operator, `.`, `/`, `+`, or `-`, it will need two operands. Pop the `operand_stack` twice. The first pop is the second operand and the second pop is the first operand. Perform the arithmetic operation. Push the result back on the `operand_stack`.
4. When the input expression has been completely processed, the result is on the stack. Pop the `operand_stack` and return the value.

To assist with the arithmetic, a helper function `do_math()` is defined:

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To assist with the arithmetic, a helper function `doMath()` is defined:

```
double doMath(string op, double op1, double op2) {  
    if (op == "*") {  
        return op1 * op2;  
    } else if (op == "/") {  
        return op1 / op2;  
    } else if (op == "+") {  
        return op1 + op2;  
    } else {  
        return op1 - op2;  
    }  
}
```

```
In [9]: #include <iostream>
#include "pythonds3/cppds/expression.hpp" // doMath + postfixEval
using namespace std;

int main() {
    cout << postfixEval("7 8 + 3 2 + /") << endl;
    return 0;
}
```

3

Expected: 3 — that is $(7 + 8) / (3 + 2)$.

Expected: 3 — that is $(7 + 8) / (3 + 2)$.

It is important to note that in both the postfix conversion and the postfix evaluation programs we assumed that there were no errors in the input expression.

In []: `display_quiz(path+"c1.json", max_width=800)`

Convert the following infix expression to postfix: $10 + 3 * 5 / (16 - 4)$

☐ $10\ 3\ 5\ *\ /\ 16\ 4\ -\ +$

☐ $10\ 3\ 5\ *\ 16\ 4\ -\ /\ +$

☐ $10\ 3\ 5\ 16\ 4\ -\ /\ * +$

☐ $10\ 3\ +\ 5\ * 16\ 4\ -\ /\$

```
In [ ]: display_quiz(path+"c2.json", max_width=800)
```

What is the result of evaluating the following postfix expression:
 $17\ 10\ +\ 3\ *\ 9\ /\ ?$

☐ 81

☐ 27

☐ 9

☐ 15

Exercise: Extend `infixToPostfix()` to convert $5 * 3 ^ (4 - 2)$.

Exercise: Extend `infixToPostfix()` to convert `5 * 3 ^ (4 - 2)`.

```
In [10]: #include <iostream>
#include <stack>
#include <vector>
#include <map>
#include <sstream>
#include <string>
#include <cctype>
using namespace std;

string infixToPostfix(string infixExpr) {
    // Your code here: add support for the right-associative ^ operator

    return result;
}

int main() {
    cout << infixToPostfix("5 * 3 ^ ( 4 - 2 )") << endl;
    return 0;
}
```

/tmp/tmp3q6wh2du.cpp: In function 'std::string infixToPostfix(std::string)':

/tmp/tmp3q6wh2du.cpp:15:12: error: 'result' was not declared in this scope

```
15 |         return result;
    |                   ^~~~~~
```

Hint: give $\wedge$ the highest precedence, and because it is **right-associative**, pop waiting operators only while their precedence is **strictly greater** than the incoming $\wedge$.

Hint: give $\wedge$ the highest precedence, and because it is **right-associative**, pop waiting operators only while their precedence is **strictly greater** than the incoming $\wedge$.

Expected: $5\ 3\ 4\ 2\ -\ \wedge\ *$, which evaluates to 45 .

3.10. Queues

A queue is an ordered collection of items where the addition of new items happens at one end, called the rear, and the removal of existing items occurs at the other end, commonly called the front. As an element enters the queue it starts at the rear and makes its way toward the front, waiting until that time when it is the next element to be removed. The item that has been in the collection the longest is at the front. This ordering principle is sometimes called FIFO, first in, first out.

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The simplest example of a queue is that we wait in a line for a movie, we wait in the checkout line at a grocery store, and we wait in the cafeteria line.

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The simplest example of a queue is that we wait in a line for a movie, we wait in the checkout line at a grocery store, and we wait in the cafeteria line.

Well-behaved lines, or queues, are very **restrictive** in that they have only one way in and only one way out. There is no jumping in the middle and no leaving before you have waited the necessary amount of time to get to the front.

Computer science also has common examples of queues. Figure shows a simple queue of data objects.

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Operating systems use a number of different queues to control processes within a computer. The scheduling of what gets done next is typically based on a queuing algorithm that tries to execute programs as quickly as possible and serve as many users as it can.

3.11 The Queue Abstract Data Type

The queue operations are given below:

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- `std::queue<T> q;` constructs an empty queue.
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- `front()` and `back()` inspect the two ends.
- `pop()` removes the front and returns `void`.
- `empty()` and `size()` inspect the queue state.

Call `front()` before `pop()` when the removed value is needed. Calling `front()`, `back()`, or `pop()` on an empty queue violates the precondition.

The conceptual table lists `front()` on the left and `back()` on the right, matching the normal STL view.

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<code>std::queue<int></code> operation	Conceptual contents (front at left)	Return value
<code>q.empty()</code>	<code>[]</code>	<code>true</code>
<code>q.push(4)</code>	<code>[4]</code>	<code>void</code>
<code>q.push(7)</code>	<code>[4, 7]</code>	<code>void</code>
<code>q.front()</code>	<code>[4, 7]</code>	<code>4</code>
<code>q.back()</code>	<code>[4, 7]</code>	<code>7</code>
<code>q.pop()</code>	<code>[7]</code>	<code>void</code>

3.12 Using `std::queue` in C++

The cppds C++ text uses `std::queue<T>`. Enqueue with `push`, inspect with `front`, then remove with `pop` (which returns `void`). `back` observes the rear. All are $O(1)$ with the default `deque` backend.

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```
#include <queue>

std::queue<std::string> q;
q.push("first");
q.push("second");
std::cout << q.front(); // first
q.pop();                // removes first; returns void
```

Course implementation supplement: `pythonds3/cppds/queue.hpp` uses `enqueue`, value-returning `dequeue`, and `isEmpty` for assignment compatibility.

```
In [11]: #include <iostream>
#include <queue>
#include <string>
using namespace std;

int main() {
    queue<string> q;
    q.push("4"); q.push("dog"); q.push("true");
    cout << q.size() << endl;
    cout << boolalpha << q.empty() << endl;
    cout << q.front() << endl;
    q.pop();
    cout << q.front() << endl;
}
```

3

false

4

dog

In []: `display_quiz(path+"queue.json", max_width=800)`

Suppose you have the following series of queue operations. What items are left on the queue?

```
std::queue<std::string> q;  
q.push("hello");  
q.push("dog");  
q.push("3");  
q.pop();
```

☐ {"3"}

☐ {"dog", "3"}

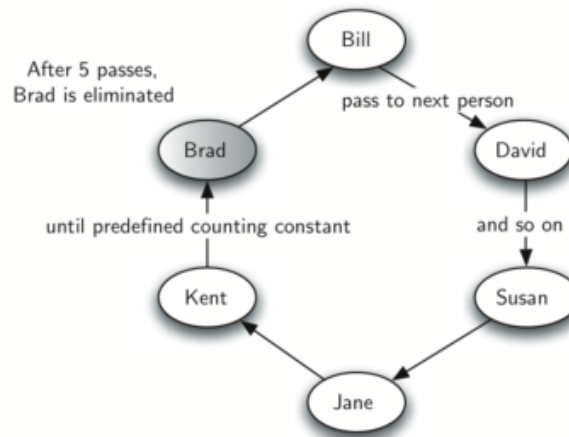
☐ {"hello", "dog", "3"}

☐ {"hello", "dog"}

3.13. Queue Simulation: Hot Potato

To begin, let's consider the children's game hot potato. In this game, children line up in a circle and pass an item from neighbor to neighbor as fast as they can. At a certain point in the game, the action is stopped and the child who has the item (the potato) is removed from the circle. Play continues until only one child is left.

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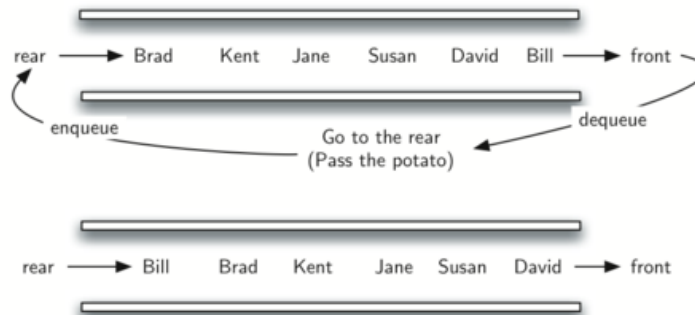
The simulation receives a `vector<string>` of names and a count. The active circle is a `std::queue<string>`; it returns the last remaining name.

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To simulate the circle, we will use a queue. Assume that the child holding the potato will be at the front of the queue. Upon passing the potato, the simulation will simply dequeue and then immediately enqueue that child, putting them at the end of the line.

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After num dequeue/enqueue operations, the child at the front will be removed permanently and another cycle will begin. This process will continue until only one name remains (the size of the queue is 1)!

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```
In [12]: #include <iostream>
#include <queue>
#include <vector>
using namespace std;

// STL queue: front() reads, pop() removes (returns nothing)
string hotPotato(vector<string> nameList, int num) {
    queue<string> simQueue;
    for (string name : nameList) simQueue.push(name);
    while (simQueue.size() > 1) {
        for (int i = 0; i < num; i++) {
            simQueue.push(simQueue.front());
            simQueue.pop();
        }
        simQueue.pop();
    }
    return simQueue.front();
}

int main() {
    cout << hotPotato({"Bill", "David", "Susan", "Jane", "Kent", "Brad"}, 7) <
    return 0;
}
```

Susan

Expected: Susan survives the game.

Expected: Susan survives the game.

Names are pushed in their given order, so the first name is initially at `front()`.
Moving one player means copying `front()`, pushing that value at the rear, and then popping the old front.

Exercise: Implement the same simulation with the course `Queue<T>` API and document the API translation.

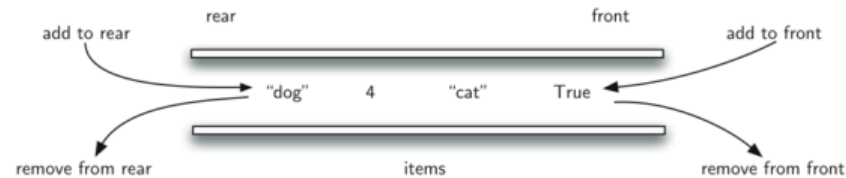
Exercise: Implement the same simulation with the course `Queue<T>` API and document the API translation.

```
In [13]: // Your code here
```

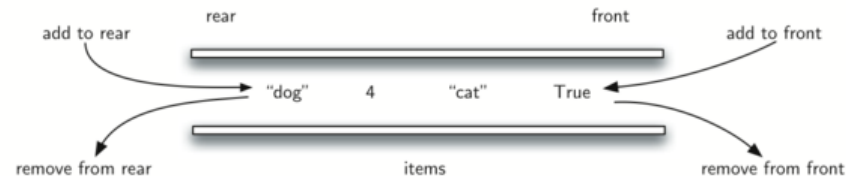
3.15 Deques

A deque, also known as a double-ended queue, is an ordered collection of items similar to the queue. It has two ends, a front and a rear. What makes a deque different is the unrestrictive nature of adding and removing items.

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New items can be added at either the front or the rear. Likewise, existing items can be removed from either end. In a sense, this hybrid linear structure provides all the capabilities of stacks and queues in a single data structure.

3.16 The Deque Abstract Data Type

The deque operations are given below:

- `std::deque<T> d;` constructs an empty deque.
- `push_front(item)` and `push_back(item)` add at either end.

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- `front()` and `back()` inspect the ends.
- `pop_front()` and `pop_back()` remove and return `void`.
- `empty()` and `size()` inspect state.

The supplementary `python3 Deque<T>` instead uses `addFront/addRear/removeFront/removeRear/isEmpty`.

For `std::deque`, `front()` is shown on the left and `back()` on the right.

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<code>std::deque<int></code>	operation	Contents (front at left)	Return value
<code>d.empty()</code>		<code>[]</code>	<code>true</code>
<code>d.push_back(4)</code>		<code>[4]</code>	<code>void</code>
<code>d.push_front(7)</code>		<code>[7, 4]</code>	<code>void</code>
<code>d.front()</code>		<code>[7, 4]</code>	<code>7</code>
<code>d.back()</code>		<code>[7, 4]</code>	<code>4</code>
<code>d.pop_front()</code>		<code>[4]</code>	<code>void</code>

3.17 Using `std::deque` in C++

The C++ Standard Library `std::deque<T>` supports constant-time insertion and removal at both ends with `push_front`, `push_back`, `pop_front`, and `pop_back`.

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```
#include <deque>  
  
std::deque<int> d;  
d.push_front(2);  
d.push_back(7);  
int first = d.front();  
d.pop_front();  
int last = d.back();  
d.pop_back();
```

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```
#include <deque>

std::deque<int> d;
d.push_front(2);
d.push_back(7);
int first = d.front();
d.pop_front();
int last = d.back();
d.pop_back();
```

Course implementation supplement. `pythonds3/cppds/deque.hpp` uses `addFront`, `addRear`, and value-returning `removeFront` / `removeRear`. Its chosen vector representation makes the operations at one end $O(n)$; this is not the complexity of `std::deque`.

Use `std::deque` as the normal C++ implementation. Use the course `Deque<T>` only where an exercise explicitly asks you to study its representation and operation costs.

3.18 Palindrome Checker

A palindrome is a string that reads the same forward and backward, for example, radar, toot, and madam. We would like to construct an algorithm to input a string of characters and check whether it is a palindrome.

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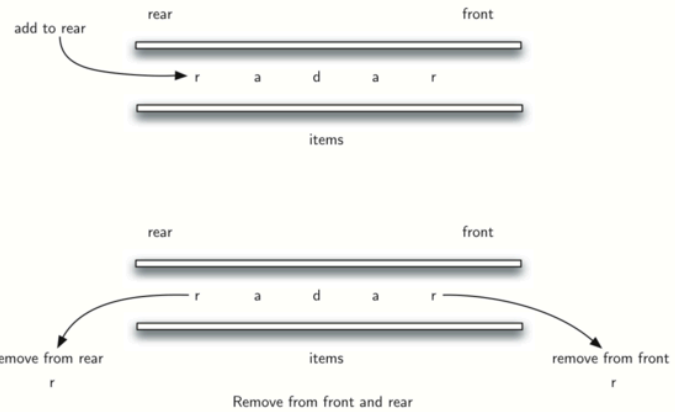
The solution to this problem will use a deque to store the characters of the string. We will process the string from left to right and add each character to the rear of the deque. At this point, the deque will be acting very much like an ordinary queue.

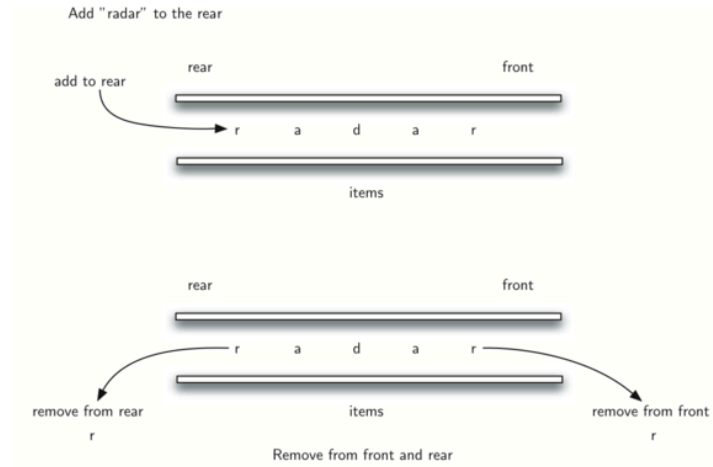
A palindrome is a string that reads the same forward and backward, for example, radar, toot, and madam. We would like to construct an algorithm to input a string of characters and check whether it is a palindrome.

The solution to this problem will use a deque to store the characters of the string. We will process the string from left to right and add each character to the rear of the deque. At this point, the deque will be acting very much like an ordinary queue.

However, we can now make use of the dual functionality of the deque. The front of the deque will hold the first character of the string and the rear of the deque will hold the last character.

Add "radar" to the rear





Since we can remove both of the front and rear characters directly, we can compare them and continue only if they match. If we can keep matching first and the last items, we will eventually either run out of characters or be left with a deque of size 1 depending on whether the length of the original string was even or odd. In either case, the string must be a palindrome.

```
In [15]: #include <iostream>
#include <deque>
#include <string>
using namespace std;

bool palChecker(string aString) {
    deque<char> charDeque;
    for (char ch : aString) charDeque.push_back(ch);    // add to the rear
    while (charDeque.size() > 1) {
        char first = charDeque.front(); charDeque.pop_front();
        char last = charDeque.back(); charDeque.pop_back();
        if (first != last) return false;
    }
    return true;
}

int main() {
    cout << boolalpha;
    cout << palChecker("lsdkjfskf") << endl;    // false
    cout << palChecker("radar") << endl;        // true
    return 0;
}
```

false

true

```
In [15]: #include <iostream>
#include <deque>
#include <string>
using namespace std;

bool palChecker(string aString) {
    deque<char> charDeque;
    for (char ch : aString) charDeque.push_back(ch);    // add to the rear
    while (charDeque.size() > 1) {
        char first = charDeque.front(); charDeque.pop_front();
        char last = charDeque.back(); charDeque.pop_back();
        if (first != last) return false;
    }
    return true;
}

int main() {
    cout << boolalpha;
    cout << palChecker("lsdkjfskf") << endl;    // false
    cout << palChecker("radar") << endl;        // true
    return 0;
}
```

false

true

Expected: false then true.

```
In [ ]: from jupytercards import display_flashcards  
display_flashcards("flashcards/ch5.json")
```

Linear ADT



Next



References

1. Textbook: *Problem Solving with Algorithms and Data Structures using C++* (cppds), Chapter 3 (Linear Structures) —
<https://runestone.academy/ns/books/published/cppds/index.html>

